Dasari Neha Vijetha

ep22btech11006@iith.ac.in | ds.nehavijetha@gmail.com | +91 9100722619 | LinkedIn

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	7.54
Narayana Junior College	Class XII (TSBIE)	2022	98.70%
Srinivas Gurukul High School	Class X (SSC)	2020	10.00%

PROFESSIONAL EXPERIENCE

AI Engineer Intern | Aurvia Technologies

- Developed AI-powered conversational workflows for analyzing business datasets and generating actionable insights.
- Worked on RAG, LLM orchestration, model routing, and analysis planning within Aurvia's AI pipeline.
- Audited the end-to-end AI chat flow from user query to response generation and identified areas for improving accuracy and context.

Automotive Tools Developer Intern | Quest Global

- Developed ARXML-to-VSS conversion using C++/Qt, enabling simplified automotive signal interoperability and standardized output.
- Implemented robust ARXML parsing to extract CAN signals, metadata, datatypes, scaling, and classify signal behavior.
- Built modular VSS YAML generation with unit normalization, hierarchical structures, and validated .vspec files.

SKILLS

Programming: Python, SQL, C++, C

AI/ML: PyTorch, TensorFlow, Scikit-learn, Hugging Face, NumPy, Pandas, SciPy

Generative AI: LLMs, RAG, Prompt Engineering, Embeddings, FAISS, ChromaDB

Tools & Deployment: FastAPI, Flask, Docker, Git, GitHub, Jupyter Notebook, Google Colab, Jira, MATLAB

PROJECTS

RAG-Based Intelligent Document Q&A System | Self Project

- Built a RAG-based PDF Q&A system using FAISS and MiniLM embeddings for accurate, source-grounded responses.
- Engineered OCR fallback, recursive chunking, and cosine-similarity retrieval to process scanned and multi-page PDFs.
- Implemented Groq/Llama 3 streaming with mandatory page-chunk citations and retrieved-source visibility to reduce hallucinations.

SplitSmart | Self Project

- Built a full-stack expense-sharing application using Flask, MySQL, Flask-SQLAlchemy, Flask-Login, and Flask-Migrate, with modular blueprints for authentication, groups, expenses, receipts, and settlements.
- Designed a greedy debt-simplification algorithm with O(n) balance calculation and O(n log n) settlement matching, reducing settlement transactions by ~90% compared with unnetted per-expense settling.
- Developed a receipt-scanning pipeline using Tesseract OCR and Gemini API to extract itemized prices, taxes, and discounts into editable forms in under 2 seconds per receipt, enabling custom item-level expense splitting.

Customer Segmentation & Image Compression using K-Means Clustering

- Built K-Means clustering pipeline using Elbow and Silhouette methods, identifying five customer segments.
- Implemented K-Means from scratch using NumPy, optimizing centroid assignment and WCSS across 900K+ points.

- Applied K-Means for image compression, reducing 16M+ colors to 2-32 representative centroids.

RELEVANT COURSES

Computer Science: Introduction to Programming (C language), Intro to AI Data structures & its Applications

Mathematics: Calculus, Probability, Statistics, Elementary Linear Algebra

Physics: Computational Physics Using Python, Data Science Analysis

Artificial Intelligence : Agentic AI (3-month course covering LangChain, LangGraph, RAG, and multi-agent systems)

SCHOLASTIC ACHIEVEMENTS

- Secured All India Rank 10 in the AMTI National Mathematics Olympiad, Sub-Junior Level.
- Participated in AI for Good 2.0 (Organised by Blend360) and Build for India AI Hackathon (Organised by Paytm)

POSITIONS OF RESPONSIBILITY

Media & PR Coordinator | ACM Student Chapter, IIT Hyderabad

- Led PR campaigns with cross-functional teams, significantly boosting event visibility and participant turnout.

Events Coordinator | Milan, IIT Hyderabad

- Coordinated logistics and cross-functional teams for events serving participants, delivering seamless execution.

Publicity Volunteer | Elan & Nvision, IIT Hyderabad

- Drove digital outreach and publicity strategy across campus channels, boosting participation for fest.

EXTRACURRICULAR

- Represented IIT Hyderabad in Inter-IIT Cultural Meet 5.0 (Dramatics) and 8.0 (Culinary) contingents, showcasing versatility
- Actively Volunteered in college-organized National Service Scheme (NSS) activities.